



Society of Petroleum Engineers

SPE-230008-MS

A Unique Case Study of Low-Resistivity Pay Development Without Fluid Sampling

Fazeel Ahmad, Dr. Ali Ahmed Jalalh, Simone Bertoli, and Hamad Rashed Khadem Al Shehhi, Al Yasat Petroleum Company Limited

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/230008-MS](https://doi.org/10.2118/230008-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

This paper presents a unique case study from a carbonate reservoir where a low-resistivity layer, situated between conventional pay zones, was successfully developed without fluid sampling. The objective is to demonstrate how integrated data interpretation—using openhole logs, pressure points, mobility, core data, and NMR—enabled confident completion decisions in ambiguous zones, resulting in dry oil production. The study aims to guide similar reservoir development strategies under uncertainty.

A multidisciplinary team conducted a comprehensive petrophysical evaluation of the low-resistivity interval using a suite of logging and core data. Openhole logs were analyzed alongside formation pressure and mobility measurements to assess fluid distribution and reservoir continuity. Core data provided insights into lithology and porosity-permeability relationships, while NMR logs were used to distinguish bound and free fluids. The team applied a cross-disciplinary interpretation workflow to differentiate conductive lithologies from water-bearing zones. Despite the absence of fluid sampling, the integrated analysis supported a decision to perforate the interval across multiple wells, guided by confidence in the data convergence.

The completion of the low-resistivity interval in several wells led to the successful production of dry oil, validating the integrated interpretation approach. Initial log responses had suggested water-bearing characteristics, but further analysis revealed that the low resistivity was due to lithological effects—such as dispersed clays and conductive minerals—rather than high water saturation. NMR data confirmed the presence of movable hydrocarbons, and pressure data supported reservoir continuity with adjacent productive zones. The absence of water production post-completion confirmed the accuracy of the interpretation and the viability of bypassed pay.

This case demonstrates that, under the right conditions, fluid sampling may not be necessary to make confident development decisions. The results highlight the value of combining multiple data sources to reduce uncertainty and unlock hidden reserves. The approach also minimized operational costs and risks associated with sampling in complex reservoirs. This case sets a precedent for similar developments in fields where low-resistivity zones are often overlooked due to conventional interpretation limitations.

Introduction

Low resistivity pay zones in carbonate reservoirs have historically posed significant challenges in hydrocarbon identification and development. These zones often exhibit ambiguous log responses that mimic water-bearing intervals, leading to bypassed pay and underutilized reserves. Conventional workflows rely heavily on fluid sampling to confirm hydrocarbon presence, which can be costly, time-consuming, and operationally risky—especially in complex reservoirs.

This paper presents a unique case study from Field X producing from Reservoir X, where a low-resistivity interval was successfully developed across three wells—Well A, Well B, and Well C—without fluid sampling. The success was achieved through a robust, integrated interpretation workflow combining petrophysical, geological, and reservoir engineering data. The results challenge conventional paradigms and offer a replicable framework for similar reservoirs worldwide.

Geological Background

Low-resistivity pay in carbonate reservoirs occurs when the rock's electrical properties are dominated by conductive lithological features rather than by movable water. In Abu Dhabi's Upper Jurassic Arab Formation - a carbonate sequence of limestones, dolomites, and interbedded anhydrites - certain sedimentological characteristics can cause resistivity logs to read abnormally low (water-like) even when the pores contain oil. Initial log analysis may thus misleadingly suggest a water-bearing zone. The key lithological factors are:

- Dispersed clay minerals in the carbonate matrix (or along bedding/stylolites), which provide additional conductive pathways via surface ionic conduction.
- Conductive minerals (notably pyrite, an iron sulfide) disseminated through the rock, which behave almost like tiny electrodes and dramatically lower the measured resistivity.
- Microporosity with bound saline water, a common feature of Middle Eastern carbonates, effectively creates an "invisible" conductive reservoir of immobile brine within the rock fabric.

These factors often coexist in the formation under study and similar carbonate reservoirs, masking hydrocarbon presence.

The Reservoir X (Upper Jurassic, Kimmeridgian) in Abu Dhabi consists of a cyclic shallow-marine carbonate platform sequence. Lithologically it alternates between high-energy grainstones/packstones (often oolitic or skeletal) and low-energy mudstone/wackestones with occasional thin shales or marl stringers (F.N. Sadooni et al, 2023). Dolomitization is common in parts, and layer-caking by anhydrite occurs at cycle tops, reflecting sabkha evaporitic conditions.

Depositional Clay: while predominantly carbonate, certain intervals (especially in the more basinal or low-energy facies) contain minor clay content. These clays can be present as:

- Argillaceous lime mudstone - the quiet water lagoonal or tidal-flat facies sometimes included fine silt and clay (e.g. windblown dust or fluvial input) that became mixed with micrite. Petrographic studies found micropores occurring "between the grains or the plates of clay minerals" in mud-supported facies¹. In other words, clay flakes are part of the rock fabric, creating micro-laminations or peloidal mud with clay skins.
- Stylolite residues - pressure solution features (stylolites) are prevalent in the Reservoir X (due to burial and compaction of this thick platform sequence). Stylolites concentrate insoluble residues, notably clays and organics, as jagged seams. These clay- rich stylolites often cut across the carbonate matrix. They can disperse clay and organic matter into the adjacent pore network or create laterally extensive conductive layers.

- Allogenic clay layers - thin shaly laminations or marly partings occasionally occur, often less than a few centimeters thick. These may represent transgressive surfaces or distal influxes of fine siliciclastics. Though thin, they can be below log resolution and effectively act as "dispersed clay" within a larger interval from a logging perspective.

Diagenetic Pyrite: The Reservoir X was subject to periods of anoxia and sulfate reduction (especially in the deeper lagoon/basin parts of Arab-D). Consequently, pyrite (FeS_2) is found as framboidal pyrite in micrite, stylolite-related pyrite and pyrite replacing fossils/organics. From a sedimentology standpoint, the Reservoir X can locally possess a hybrid carbonate-siliciclastic character: predominantly carbonate but with small fractions of clay and pyrite distributed through certain facies and diagenetic features. These pyritized carbonates have metallic conduction through pyrite networks, which can overwhelm pore fluid effects. These might be localized but when present, it causes drastic resistivity suppression.

Microporosity and Bound Water

A hallmark of Abu Dhabi's carbonate reservoirs is their extensive microporosity. This refers to pore throats in the micron to sub-micron range within micritic matrix or diagenetic cement. The Reservoir X contains microporous limestones - for instance, oomoldic grainstones with microporous ooid cortices, or mud-rich wackestones with intercrystalline micropores.

Microporosity is relevant because it usually retains capillary-bound brine that does not flow (hence does not produce water) but does conduct electricity. Essentially, the rock behaves as if it has conductive "wet sponge" filling the space between larger pores. Several studies have quantified this:

- [Hassan and Kerans \(2013\)](#) reported that in a Lower Cretaceous Abu Dhabi limestone (analogous to Arab formation fabric), microporosity made up ~70% of total porosity, and the micropores were filled with brine forming a continuous conductive network that "masks the presence of hydrocarbons" ([Hassan et al., 2013](#)).
- [Obeidi et al. \(ADNOC, 2010\)](#) describe these zones as "micro-occluded porosity" holding irreducible hyper-saline water. Even when oil occupies the macro-pores, the micropores remain 100% brine-filled and connected, yielding resistivities as low as 0.3-0.5 $\Omega\cdot\text{m}$ in an oil zone. In their wells, conventional logs calculated 80-100% water saturation in such intervals, yet DSTs produced entirely oil - a classic low-resistivity pay scenario ([Obeidi et al., 2010](#)).
- NMR logs run in these intervals show a bimodal T_2 distribution: a large volume of short- T_2 signal (bound water in micropores) and a smaller long- T_2 component (free fluid in macropores). This confirms that considerable water is present but immobile. The resistivity log, being sensitive to any connected water, sees that bound water and reads low.

Sedimentologically, microporosity in the Reservoir X originates from micritization of grains and early diagenesis. Many grains (ooids, bioclasts) in the Arab carbonates are enveloped by micrite or have been partially dissolved leaving micro-crystalline matrices. Additionally, the matrix of mud-supported layers is microcrystalline calcite. During the burial, compaction and minor cementation did not eliminate these micropores. Instead, processes like grain leaching and dolomitization can even enhance microporosity. For example, it has been noticed that partial dissolution of peloids and precipitation of early cements created pervasive microporosity, later moderately occluded by calcite overgrowths and bitumen, but much still remains ([Alabere et al., 2023](#)).

In summary, the Reservoir X microporous fabric, saturated with connate brine, is a major cause of low resistivity readings. It is a lithological effect (a product of depositional texture and diagenesis) rather than an indication of high movable water. The presence of dispersed clays often goes hand-in-hand with microporosity - clay particles reside in those micropores or along their surfaces, further enhancing

conductivity. Thus, the "dispersed clays" and "micropore bound water" work in tandem to depress resistivity in these rocks.

Field Overview

Field X is a carbonate reservoir located in Offshore Abu Dhabi, under its 1st development phase, characterized by:

- **Depositional Environment:** Shallow marine platform carbonates with interbedded micritic limestone and dolomite.
- **Reservoir Characteristics:** Moderate porosity (10-20%), low to medium permeability (10-50 mD in matrix), and presence of natural fractures.
- **Challenges:** Complex pore structures contribute to low-resistivity responses in certain intervals.

Objectives

The original development focuses on high-resistivity zones with confirmed oil saturation. However, recent re-evaluation of low-resistivity intervals revealed additional opportunity, prompting a multidisciplinary team to investigate with the following objectives:

- To evaluate the hydrocarbon potential of a low-resistivity interval situated between conventional pay zones.
- To demonstrate the effectiveness of integrated data interpretation in the absence of fluid sampling.
- To validate the approach through successful dry oil production from multiple wells.
- To establish a workflow that can be replicated in similar reservoirs globally.

Data and Methodology

The evaluation and development of the low-resistivity interval in Field X were carried out through a structured, multidisciplinary workflow. The methodology integrated petrophysical, geological, and reservoir engineering data to overcome the ambiguity typically associated with low-resistivity zones.

Data Acquisition

The following datasets were acquired across Well A, Well B, and Well C:

- **Openhole logs:** Gamma ray, LWD Multi propagation resistivity (deep, medium, shallow), density, neutron, and photoelectric factor (PEF).
- **NMR logs:** Used to differentiate bound water from free fluids and to estimate total and effective porosity.
- **Formation pressure and mobility data:** Acquired via LWD Formation Dynamics Tester to assess reservoir continuity and fluid mobility.
- **Core data:** available in the nearby wells, including plug measurements for porosity, permeability, grain density, and thin section analysis for lithofacies classification.
- **Production data:** Post-ICD completion production behavior.

Petrophysical Evaluation

Carbonate reservoirs are complex and heterogeneous, making it challenging to identify and characterize hydrocarbon-bearing zones. Low resistivity pay zones, in particular, are difficult to detect using traditional logging methods, such as resistivity and porosity logs. These zones are characterized by low resistivity

values, often similar to those of water-bearing zones, making it difficult to distinguish between hydrocarbon-bearing and water-bearing intervals.

Low resistivity pay zones present a unique challenge in petrophysical evaluation, often leading to underestimation of hydrocarbon potential. A comprehensive approach integrating resistivity-porosity cross-plots, T2 distribution analysis, and pressure data can help resolve ambiguities and improve reservoir characterization.

In resistivity-porosity cross-plots, low resistivity pay typically deviates from the expected trend of clean hydrocarbon-bearing formations. This anomaly may be attributed to conductive minerals (e.g., clays), high water salinity, or complex pore structures. By overlaying core-calibrated saturation models, these plots can help significantly to differentiate between true water-bearing zones and hydrocarbon-bearing intervals with suppressed resistivity.

Resistivity Log response. The resistivity log values for the water bearing zones range from 0.22 to 1.1 Qm, depending on its formation factor. In the oil leg, RT values range from 0.3 to 1.4 Qm. Water formation is brine, with salinities of 200 Kppm NaCl eq. and hence, RT log, regarding oil interval, RT values are close to the water bearing RT signature, making challenging the accurate assessment for the oil. The Figure-3 shows some examples for oil interval and water bearing of RT log.

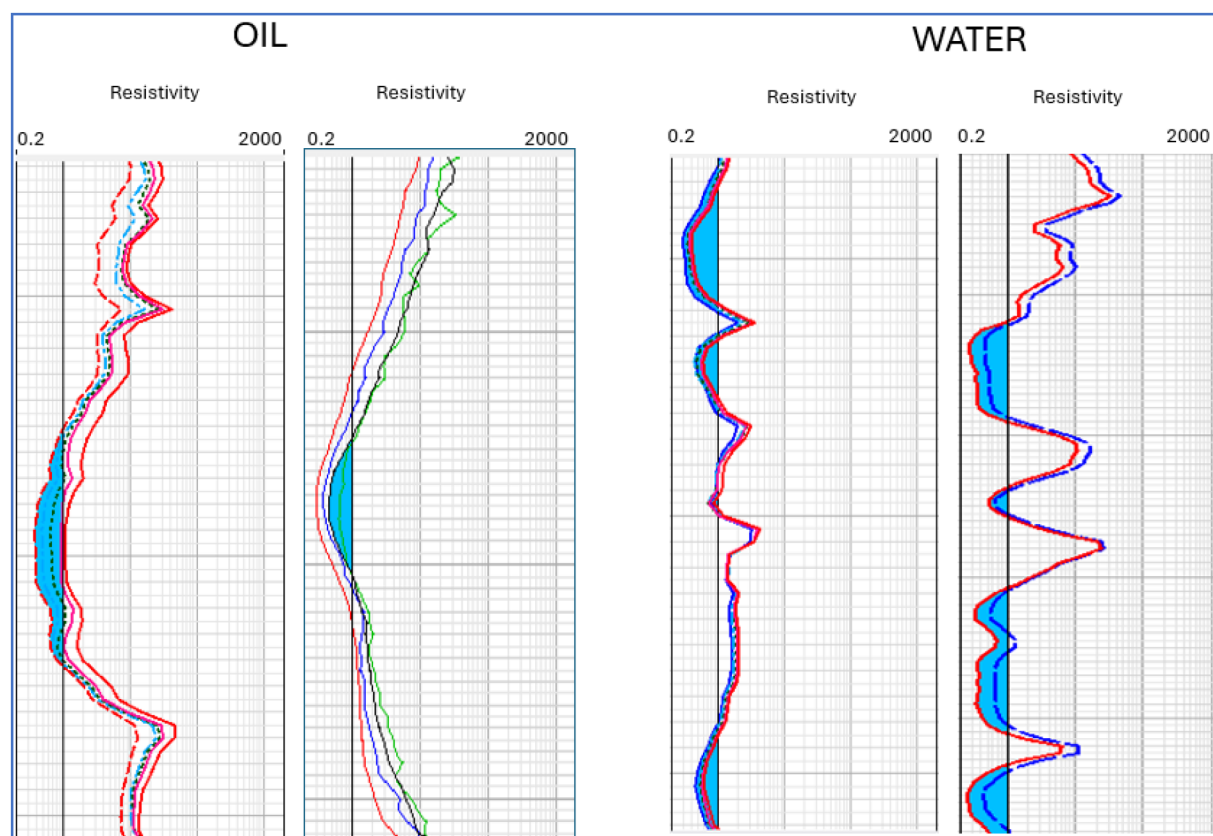


Figure 1—RT flag examples for Oil Interval (left side) and Water (right side). Depths on feet MD. RT Base line set at 1.0 ohmm

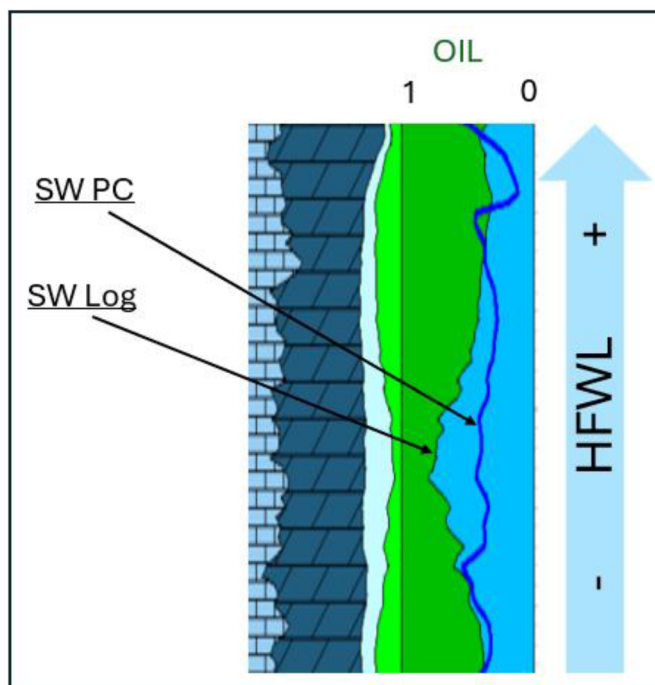


Figure 2—Showing Saturation derived from log and MICP core data.

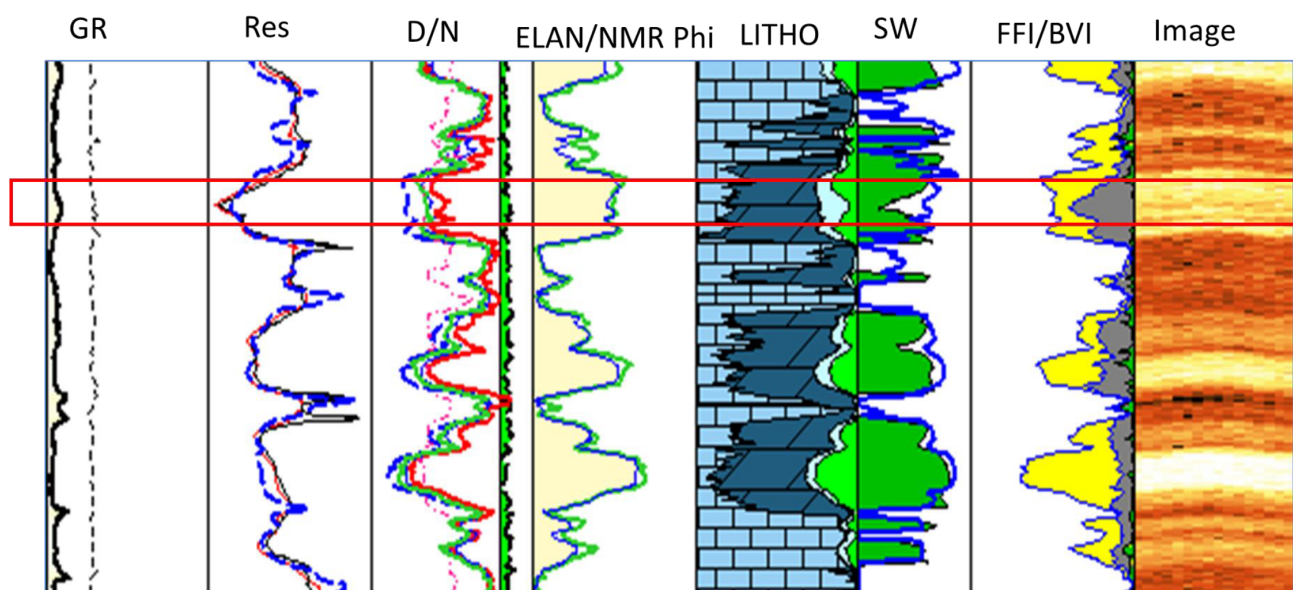


Figure 3—Well A: Openhole Logs Evaluation

Logs vs. SCAL Water Saturations: When comparing SW from both logs and PC-SCAL, it is observed a consistent mismatch in the oil interval resulting in higher water content calculated from the log. This disagreement is only noted in the oil interval. Figure-4, illustrates these differences in saturations for both oil intervals with low resistivity pay and without microporosity effect, showing that the SW log derived increases as going higher in the oil column, while SW PC derived decreases in the oil interval, when considering same Petrophysical Group.

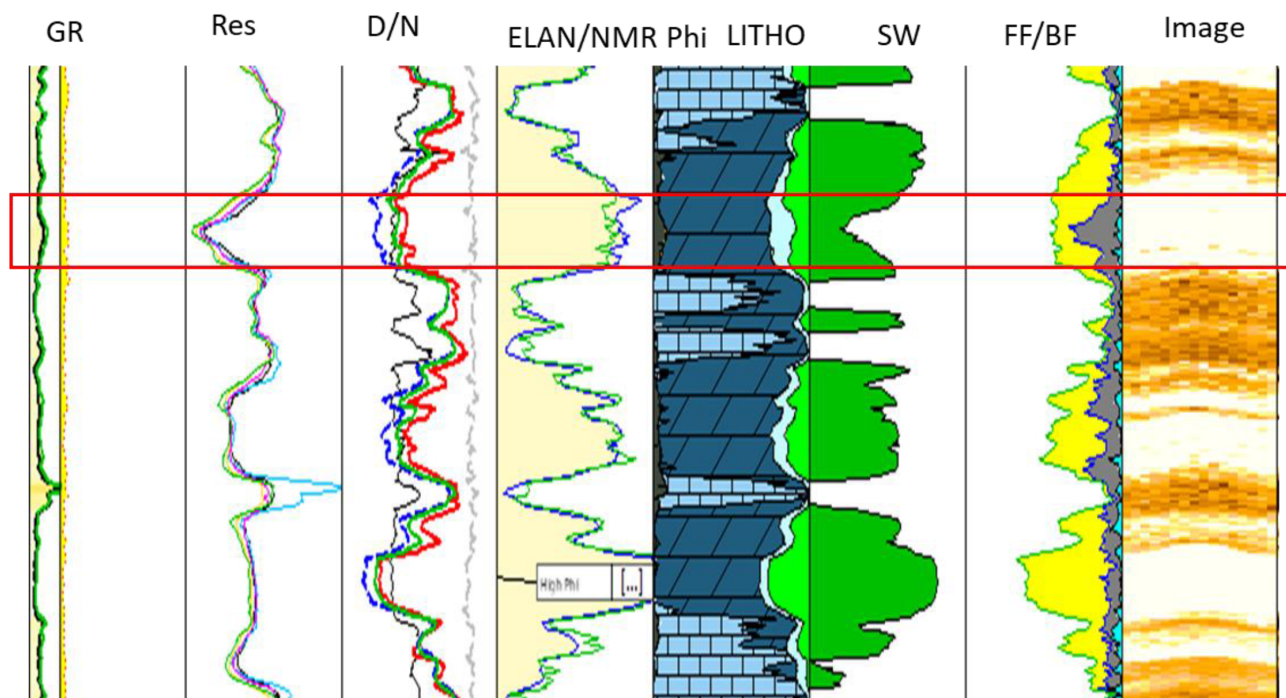


Figure 4—Well B: Openhole Logs Evaluation

NMR T2 distribution analysis adds another layer of insight by characterizing pore size distribution and fluid types. In low resistivity pay, T_2 spectra often show a bimodal distribution, indicating the presence of both bound and free fluids. The presence of longer T_2 components, especially when supported by high porosity and low irreducible water saturation, suggests movable hydrocarbons despite low resistivity readings.

T2 distribution analysis is a key component of Nuclear Magnetic Resonance (NMR) logging, used to evaluate pore structure and fluid types in reservoir rocks. Here's a detailed explanation:

T_2 , or **transverse relaxation time**, measures how quickly hydrogen nuclei in fluids lose phase coherence after being excited by a magnetic field. In porous media, this relaxation is influenced by:

- **Pore size**
- **Surface-to-volume ratio**
- **Fluid type and viscosity**

Shorter T_2 times typically indicate smaller pores or bound fluids, while longer T_2 times suggest larger pores or free fluids.

NMR T2 distribution analysis adds another layer of insight by characterizing pore size distribution and fluid types. In low resistivity pay, T_2 spectra often show a bimodal distribution, indicating the presence of both bound and free fluids. The presence of longer T_2 components, especially when supported by high porosity and low irreducible water saturation, suggests movable hydrocarbons despite low resistivity readings.

T2 distribution analysis is a key component of Nuclear Magnetic Resonance (NMR) logging, used to evaluate pore structure and fluid types in reservoir rocks. Here's a detailed explanation:

T_2 , or **transverse relaxation time**, measures how quickly hydrogen nuclei in fluids lose phase coherence after being excited by a magnetic field. In porous media, this relaxation is influenced by:

- **Pore size**

- **Surface-to-volume ratio**
- **Fluid type and viscosity**

Shorter T2 times typically indicate smaller pores or bound fluids, while longer T2 times suggest larger pores or free fluids.

T2 Distribution Curve

The T2 distribution is a histogram showing the volume of fluid associated with different relaxation times. It provides a **quantitative pore size distribution** and helps distinguish between:

- **Clay-bound water (CBW):** Very short T2 times (<3 ms), immobile.
- **Capillary-bound water (BVI):** Intermediate T2 times (3-92 ms), mostly immobile.
- **Free fluid (FFI):** Long T2 times (>92 ms), potentially producible hydrocarbons.

These cutoffs vary by formation and are often calibrated with core data.

Applications in Low Resistivity Pay

In low resistivity pay zones, conventional logs may misinterpret hydrocarbon-bearing intervals such as water-bearing due to suppressed resistivity. T2 distribution helps resolve this by:

- **Identifying movable hydrocarbons:** Presence of long T2 components indicates free fluids, even when resistivity is low.
- **Quantifying bound vs. free fluids:** Differentiates between clay-bound water and producible oil.
- **Supporting saturation models:** Enhances interpretation when combined with resistivity and porosity data.

NMR Interpretation

Carbonate reservoirs are complex and heterogeneous, making it challenging to identify and characterize hydrocarbon-bearing zones. Low resistivity pay zones are difficult to detect using traditional logging methods, such as resistivity and porosity logs. These zones are characterized by low resistivity values, often similar to those of water-bearing zones, making it difficult to distinguish between hydrocarbon-bearing and water-bearing intervals.

NMR Logging in Low Resistivity Pay

NMR logging has emerged as a valuable tool in identifying and characterizing low resistivity pay in carbonate reservoirs. NMR logs provide information on the pore size distribution, porosity, and permeability of the reservoir, which can be used to identify hydrocarbonbearing zones. The NMR log response is sensitive to the presence of hydrocarbons in the pore space, allowing for the identification of low resistivity pay zones.

- **T2 Distribution Analysis:** Provided insights into pore size distribution and fluid types. Movable hydrocarbons were identified based on T2 cutoff values.
- **BVI vs. FFI Mapping:** Bound Volume Index (BVI) and Free Fluid Index (FFI) were used to distinguish producible oil from bound water.
- **Total vs. Effective Porosity:** NMR-derived porosity was compared with conventional logs to validate reservoir quality.

Benefits and Limitations of NMR Logging

The benefits of NMR logging in low resistivity pay zones of carbonate reservoirs include:

1. **Improved identification of hydrocarbon-bearing zones:** NMR logging can identify low resistivity pay zones that may be missed by traditional logging methods.
2. **Enhanced reservoir characterization:** NMR logs provide information on the pore size distribution, porosity, and permeability of the reservoir, which can be used to optimize production and injection strategies. However, NMR logging also has some limitations, including:
3. **Depth of investigation:** NMR logs have a limited depth of investigation, which can affect the accuracy of the measurements in cases where the reservoir is highly heterogeneous.
4. **Interpretation challenges:** NMR log interpretation requires expertise and can be challenging in complex carbonate reservoirs.

Pressure and Mobility Integration

- **Pressure Analysis:** The measured pressures response indicates the low resistivity zone is connected with the top high saturation zone
- **Mobility Trends:** Mobility values were used to infer permeability and validate producibility, especially in zones with ambiguous log responses.

Multidisciplinary Integration

- **Collaborative Interpretation:** Petrophysicist, geologists, and reservoir engineers jointly reviewed data to ensure consistency and reduce uncertainty.
- **Decision Matrix:** A confidence-based decision matrix was developed to guide ICD completion decisions, incorporating data convergence from multiple sources.
- **Risk Assessment:** Operational and reservoir risks were evaluated, including the potential for water breakthrough and misinterpretation of conductive zones.

Production Completion Strategy

- **Selective ICD completion:** Based on integrated interpretation, ICD intervals were selected to target zones with confirmed movable hydrocarbons and reservoir continuity.
- **Operational Execution:** ICD completion was ran successfully as per the design

Results and Discussion

Well A

Well A is the 1st development well in the field with extensive data gathering program including NMR Log, pressure & mobility data which were critical in confirming movable hydrocarbons. [Figure-3](#) shows the consolidated log interpretation section across the target section of the well. The well was completed with ICD's and post-completion total oil rate tested was 3,500 BOPD, water cut: 0%. The individual contribution of low resistivity pay interval is not available, however the zero-water cut confirmed the understanding of no moveable water from low resistivity pay which provides confidence about the methodology of assessing such challenging interval for completion.

Well B

Well B is the 2nd development well in the field with extensive data gathering program including NMR Log, pressure & mobility data which were critical in confirming movable hydrocarbons. [Figure-4](#) shows the

consolidated log interpretation section across the target section of the well. The well was completed with ICD's and post-completion total oil rate tested was 3,000 BOPD, water cut: 0%. The individual contribution of low resistivity pay interval is not available, however the zero-water cut confirmed the understanding of no moveable water from low resistivity pay which provides confidence about the methodology of assessing such challenging interval for completion.

Well C

Well C also produced dry oil from the low-resistivity interval. No NMR log was acquired because the LRP resistivity signature responses were already confirmed from Well A and B. Formation pressure & mobility data were critical and acquired in confirming the reservoir connectivity. The resistivity response of LRP layer in Well C, is the same response observed in Well A and B as shown in [figure-5](#). Post-completion total oil rate tested ~3200 BOPD, water cut: 0%. The individual contribution of low resistivity pay interval is not available, however the zero-water cut confirmed which provides confidence about the methodology of assessing such challenging interval for completion.

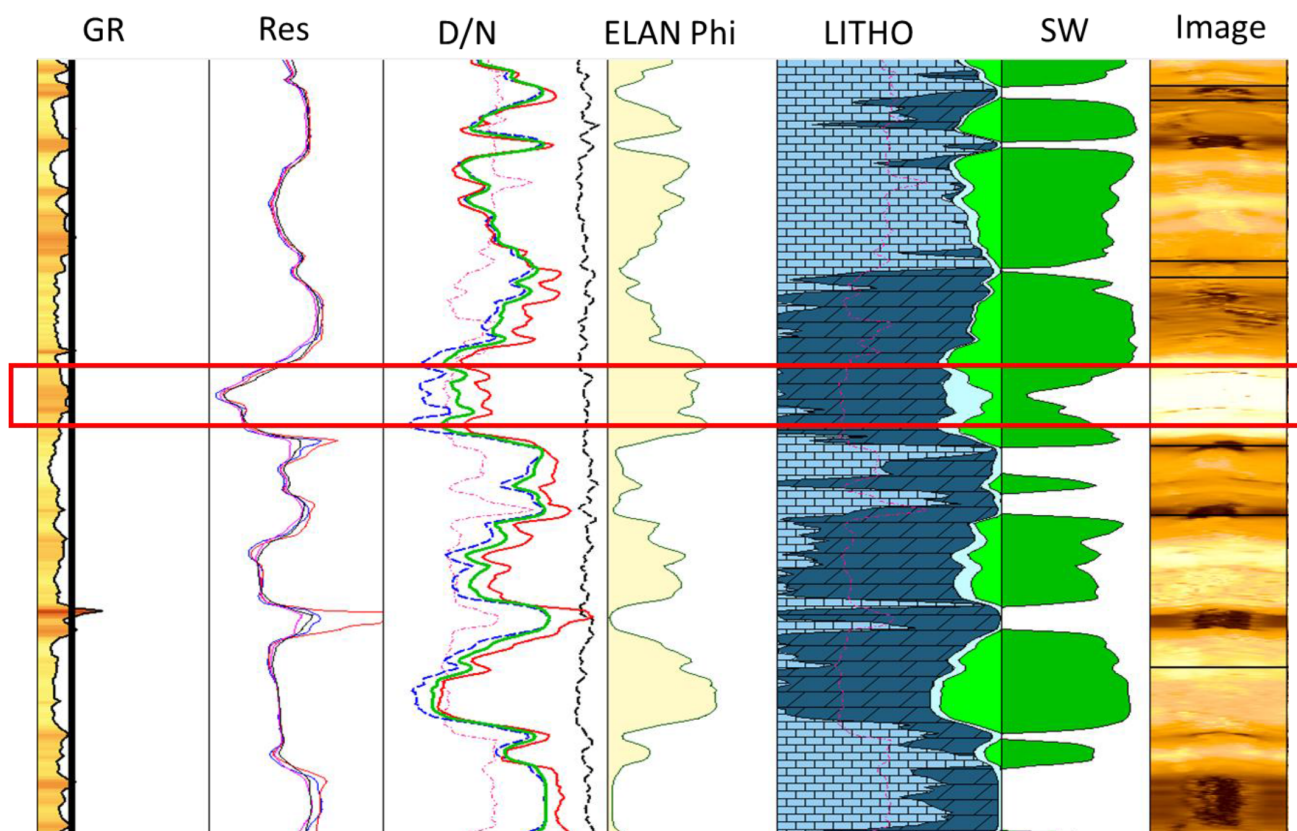


Figure 5—Well C: Openhole Logs Evaluation

Conclusions

- Low resistivity pay zones in carbonate reservoirs can be challenging to identify using traditional logging methods.
- NMR logging has emerged as a valuable tool in identifying and characterizing these zones. The benefits of NMR logging include improved identification of hydrocarbon-bearing zones and enhanced reservoir characterization. However, NMR logging also has some limitations, including limited depth of investigation and interpretation challenges. By understanding the benefits and

limitations of NMR logging, operators can optimize their logging and production strategies to maximize hydrocarbon recovery from low resistivity pay zones in carbonate reservoirs.

- Lithology and diagenesis impart conductive features in the Arab Formation; knowing the depositional facies and diagenetic history allows us to predict where low resistivity pay might occur.
- The low resistivity can be driven by clay-bearing carbonates, microporous carbonates with fine-grained facies and pyritized carbonates.
- Low-resistivity intervals can contain producible hydrocarbons if properly evaluated.
- Integrated interpretation using NMR, MDT, core, and openhole logs can de-risk development decisions.
- Fluid sampling may not be necessary under high-confidence data convergence.
- The approach minimized operational costs and unlocked an additional opportunity for development.
- This methodology is replicable in similar carbonate reservoirs globally.

Acknowledgments

The authors would like to thank ADNOC Upstream, CNPC, and Al Yasat Petroleum for their support. Special thanks to the multidisciplinary team for their collaboration & contributions.

References

- Sadooni, F.N., Al-Kuwari, H.A.S., Sakhaee-Pour, A. et al (2023). Microporosity evolution and destruction in the Jurassic Arab D reservoir, Qatar. *Carbonates Evaporites* **38**, 45.
- Hassan, A. & Kerans, C. (2013). Rock fabric characterization in a low-resistivity pay carbonate, Abu Dhabi. SEG Technical Program Expanded Abstracts, 2932-2936
- Obeidi, A., Al Aryani, F., Al Amoudi, M. (2010). Developed Approach for Better Understanding of Low Resistivity Pay Carbonate Reservoirs. SPE 137663
- Alabere, A., Alsuwaidi, M., Hassan, A. A., Mansurbeg, H., Morad, S., Al-Shalabi, E. W., & Jallad, O. A. (2023). Controls on the formation and evolution of multimodal pore network in lower cretaceous limestone reservoir, Abu Dhabi, United Arab Emirates. *Marine and Petroleum Geology*, 152, Article 106222.
- Coates, G. R., et al (1999). "A new approach to improved log-derived permeability." *SPE Reservoir Evaluation & Engineering*, **2**(04), 382–391.
- Kenyon, W. E. (1997). "Petrophysical principles of applications of NMR logging." *The Log Analyst*, **38**(02), 21–43.